

Appendix 2

a_2 as a function of detector width δ_1

Calculating a_2^{α} using Numerical Integration

$$k_0 = 20;$$

$$X_0 = -5;$$

$$X_d = -3;$$

$$\delta = 0.5;$$

$$T_i = -0.02;$$

$$T_f = 0.22;$$

$$dt = (T_f - T_i) / (n - 1);$$

$$tm[0] = T_i - dt;$$

$$tm[1] = T_i;$$

$$tm[n + 1] = T_f + dt;$$

$$\text{Do}[tm[i] = tm[i - 1] + dt, \{i, 2, n\}]$$

$$\text{err}[x_, T1_, T2_, D_] :=$$

$$\begin{aligned} & -\text{Erfi}\left[(-1)^{1/4} \left((-2 \pm \delta^2 + T1) (x - X_d - D/2) - T2 (X_0 + 2 \pm \delta^2 k_0 - X_d - D/2)\right) \right. \\ & \quad \left. \left(\sqrt{4 \delta^2 + 2 \pm T1} \sqrt{2 \delta^2 + \pm T1 - \pm T2} \sqrt{T2}\right)\right] / 2 + \end{aligned}$$

$$\begin{aligned} & \text{Erfi}\left[(-1)^{1/4} \left((-2 \pm \delta^2 + T1) (x - X_d + D/2) - T2 (X_0 + 2 \pm \delta^2 k_0 - X_d + D/2)\right) \right. \\ & \quad \left. \left(\sqrt{4 \delta^2 + 2 \pm T1} \sqrt{2 \delta^2 + \pm T1 - \pm T2} \sqrt{T2}\right)\right] / 2 \end{aligned}$$

$$\text{PsiStarPsi}[x_, t_] :=$$

$$\left(\frac{\delta^2}{2\pi}\right)^{1/2} * \frac{\text{Exp}[-2 k_0^2 * \delta^2]}{\sqrt{\delta^4 + \frac{t^2}{4}}} * \text{Exp}\left[\frac{(2 * \delta^2 * k_0 + \pm (x - X_0))^2}{4 * \delta^2 + 2 * \pm * t} + \frac{(2 * \delta^2 * k_0 - \pm (x - X_0))^2}{4 * \delta^2 - 2 * \pm * t}\right];$$

```

NIntegrate[PsiStarPsi[x, tm[2]] *
  (1 - err[x, tm[1], tm[2], .00005] - Conjugate[err[x, tm[1], tm[2], .00005]] +
    err[x, tm[1], tm[2], .00005] * Conjugate[err[x, tm[1], tm[2], .00005]]),
  {x, -3 - .005 / 2, -3 + .005 / 2}]

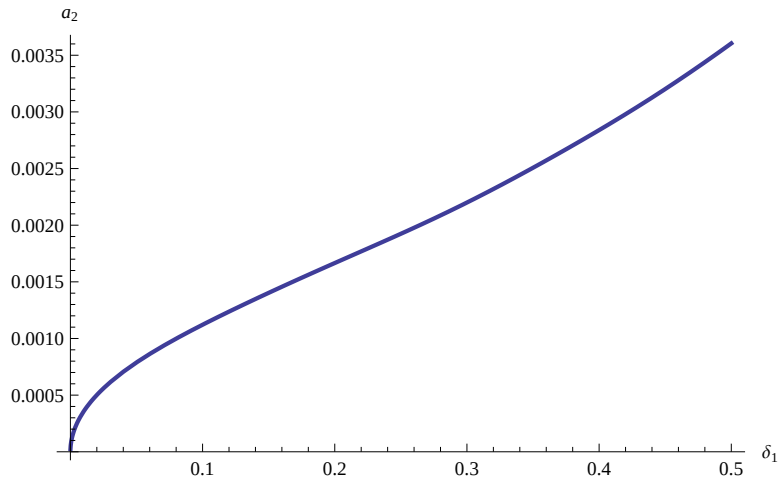
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$6.36241 \times 10^{-8} + 0. i$

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Original = Plot[(NIntegrate[PsiStarPsi[x, tm[2]] *
  (1 - err[x, tm[1], tm[2], D1] - Conjugate[err[x, tm[1], tm[2], D1]] +
    err[x, tm[1], tm[2], D1] * Conjugate[err[x, tm[1], tm[2], D1]]),
  {x, -3 - D1 / 2, -3 + D1 / 2}])^(1 / 2), {D1, 0.00005, .5},
  PlotStyle -> Thick, AxesLabel -> {" $\delta_1$ ", " $a_2$ "}]

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Calculating a^{β}_2 using Small Detector Approximation

$$\psi[0, x_, t_] := \left(\frac{\delta^2}{2\pi} \right)^{1/4} * \frac{\text{Exp}[-k_0^2 * \delta^2]}{\sqrt{\delta^2 + \frac{i * t}{2}}} * \text{Exp}\left[\frac{(2 * \delta^2 * k_0 + i * (x - X_0))^2}{4 * \delta^2 + 2 * i * t} \right]$$

$$K[xf_, tf_, xi_, ti_] := \frac{1}{\sqrt{2 * \pi * i * (tf - ti)}} * \text{Exp}\left[\frac{i * (xf - xi)^2}{2 * (tf - ti)} \right]$$

$$a[1, D_] := \text{Abs}[\psi[0, X_d, tm[1]]] * \text{Sqrt}[D];$$

$$b[1, D_] := \sqrt{1 - (a[1, D])^2};$$

$$\text{Do}[\alpha[0, j] = \psi[0, X_d, tm[j]], \{j, 1, n+1\}]$$

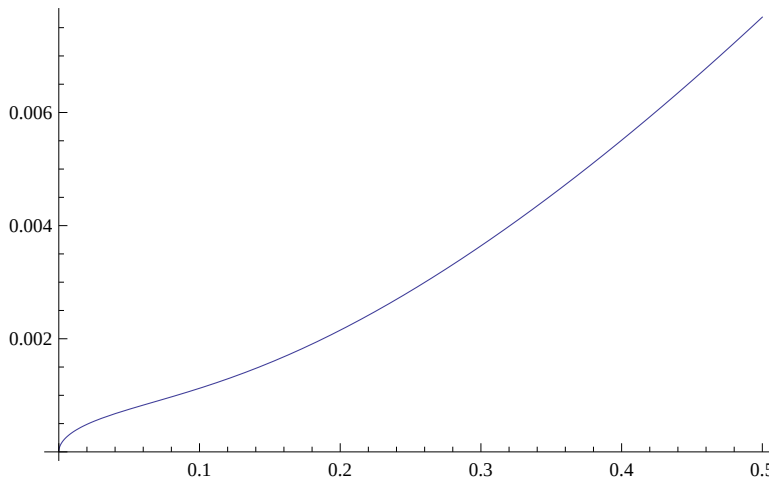
$$\text{Do}[\text{Do}[\beta[i, j] = K[X_d, 0, X_d, tm[j] - tm[i]], \{j, i+1, n+1\}], \{i, 1, n\}]$$

$$\text{Do}[\alpha[1, j] = (\alpha[0, j] - \alpha[0, 1] * \beta[1, j] * D1) / b[1, D1], \{j, 2, n+1\}];$$

$$a1[2, D1] = \text{Abs}[\alpha[1, 2]] * \text{Sqrt}[D1]$$

$$\sqrt{D1} \text{ Abs}\left[\frac{(0.00285322 - 0.00214086 i) - (0.019079 + 0.0128592 i) D1}{\sqrt{1 - 8.0633 \times 10^{-6} D1}} \right]$$

$$\text{approx} = \text{Plot}\left[\left(\sqrt{D1} \text{ Abs}\left[((0.0028532247536932768 - 0.0021408583537772564 i) - (0.019079020747766757 + 0.012859150382602592 i) D1) / \left(\sqrt{1 - 8.063295956599494 \times 10^{-6} D1} \right) \right] \right), \{D1, 0.00005, .5\} \right]$$



Calculating a^{γ}_2 using Random position Approximation

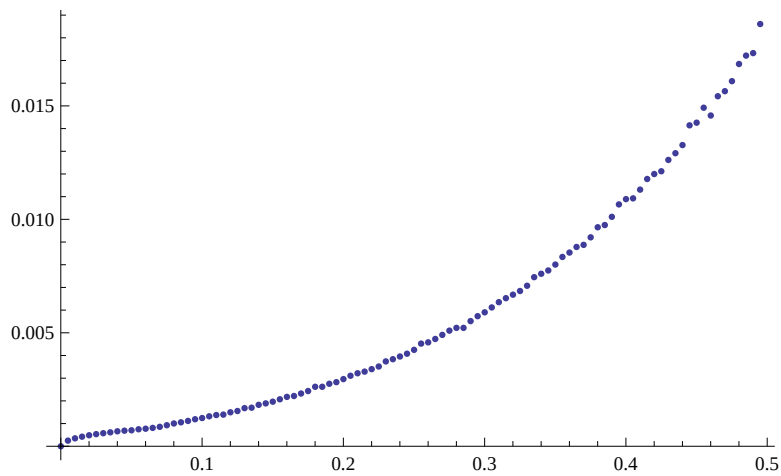
$$m = 1000;$$

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AverageList = {};
Do[{a = 0; Do[a2[1, i] = Abs[ψ[0, RandomReal[{Xd - D1, Xd + D1}], tm[1]]] * Sqrt[D1];
  b2[1, i] = √(1 - (a2[1, i])²); Do[α[0, j] = ψ[0, RandomReal[{Xd - D1, Xd + D1}], tm[j]],
    {j, 1, 2}]; Do[Do[β[i, j] = K[RandomReal[{Xd - D1, Xd + D1}], 0,
      RandomReal[{Xd - D1, Xd + D1}], tm[j] - tm[i]], {j, i + 1, 3}], {i, 1, 2}];
  Do[α[1, j] = (α[0, j] - α[0, 1] * β[1, j] * D1) / b2[1, i], {j, 2, 3}];
  a2[2, i] = Abs[α[1, 2]] * Sqrt[D1], {i, 1, 10 000 * D1}];
Val = Sum[a2[2, i], {i, 1, 10 000 * D1}], AverageList =
  Append[AverageList, {D1, (Val / (10 000 * D1))}], {D1, .00005, .5, .005}]

random = ListPlot[AverageList]

```



Comparing of the three a_2 values

Show[Original, approx, random]

